

Wetenschapsfilosofie & Statistisch Redeneren

Psychobiologie, Universiteit van Amsterdam



People

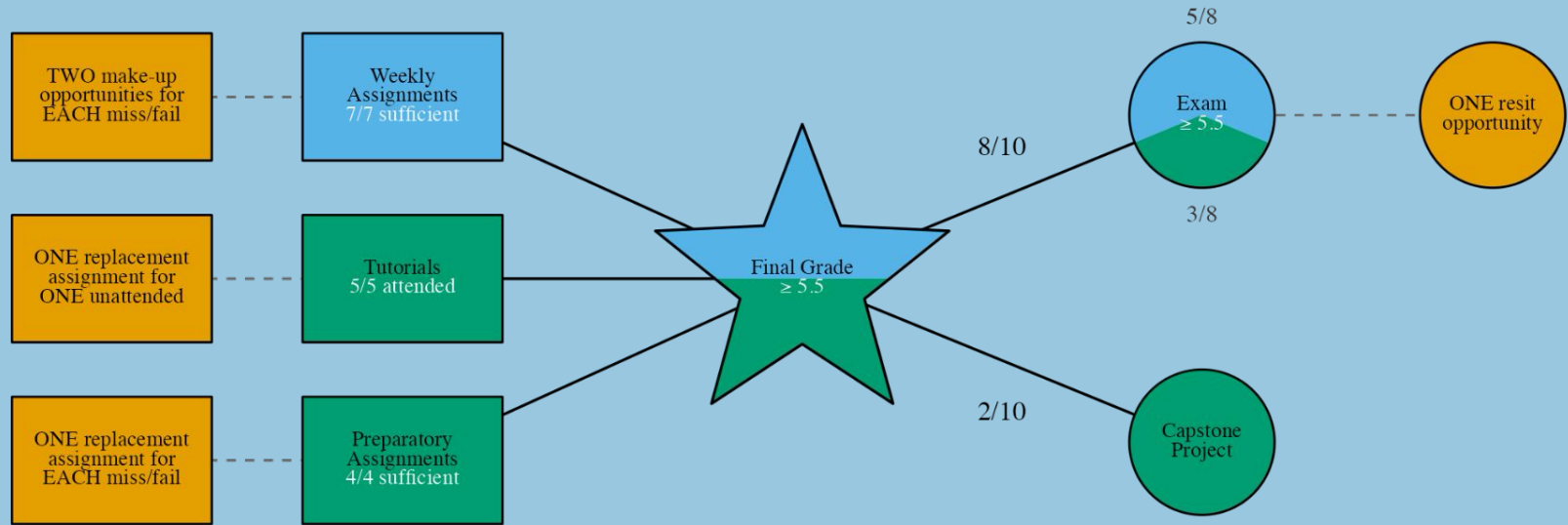
WSR WF SR 



Lecturers	Alexander Savi				Everything cross-course & substantive SR; Jonas first line of contact	Lecture/Canvas
	Jolien Francken				Substantive WF; tutorial teachers first line of contact	Lecture
Teachers	Jonas van Nijnatten				General WF; general SR; weekly assignments; assignment Q&A	Tutorial/Q&A
	Tess den Uyl				General WF	Tutorial
	Madison Carr				"	Tutorial
	Claire Paulissen				"	Tutorial
	Suzanne van Leerdam				"	Tutorial
	Jolande Disser				"	Tutorial
	Eline Hasenoot				"	Tutorial
	Lola van Linge				"	Tutorial
TAs	Maryam Shafik					-
	Nathan Pieterse					-
Education	Anne/Marleen/etc.					Email



Grading



Gates



Grades



Statistical Reasoning



Philosophy of Science



Resit

Statistical Reasoning

“The most important questions of life are, for the most part, really only problems of probability.”
— Pierre-Simon, Marquis de Laplace

 Statistical Reasoning Lecture #1
Alexander Savi, 2026

 Mehmetoglu & Mittner Ch. (1, 2, 3), 7

Curled by Danielle Navarro ([jasmynes package](#), [diy](#)) 

What are you supposed to know already?



UNIVERSITEIT
VAN AMSTERDAM



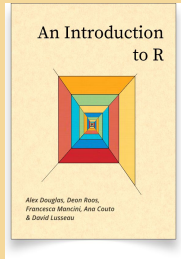
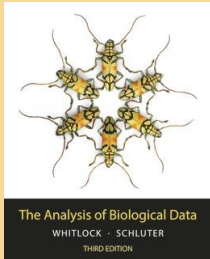
CANVAS
BY INSTRUCTURE



Studio[®]



SOWISO



Methoden van Onderzoek en Statistiek

Samples & sampling distributions

NHST

P values

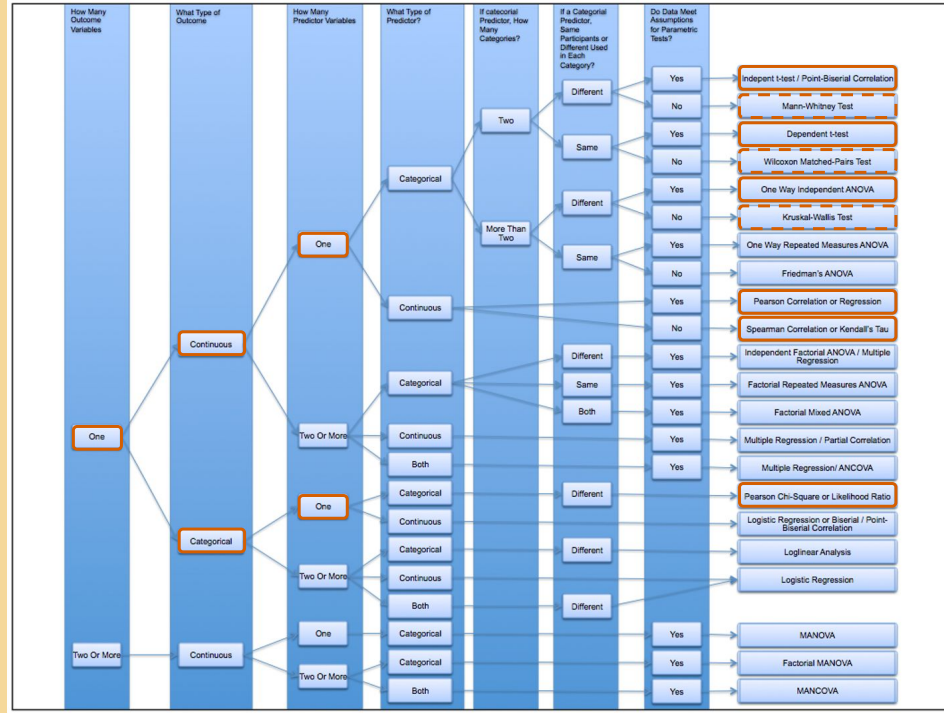
Confidence intervals

Probabilities

Bayes theorem

Test choice & research design

Data visualization



Methoden van Onderzoek en Statistiek

Binomial test

Chi-squared test

Student's t -test

One-way ANOVA

Correlation

Simple linear regression

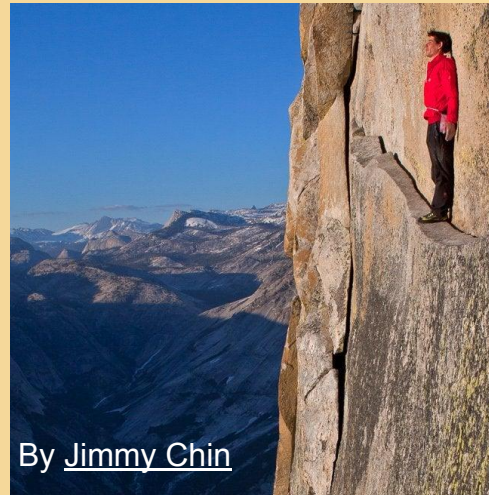
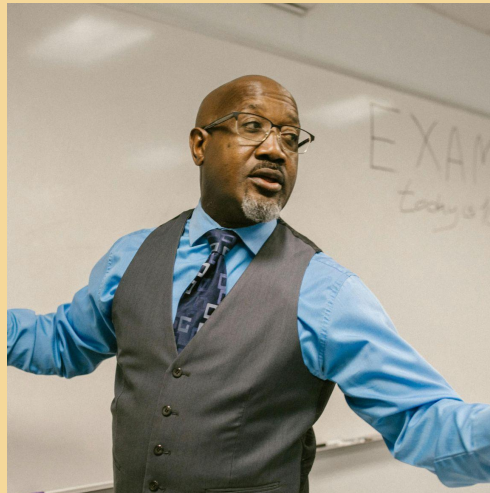
Some nonparametric alternatives

Does the unprescribed use of Ritalin (yes, no) lead to mental illnesses, such as sleeplessness and psychological dependence (yes, no)?



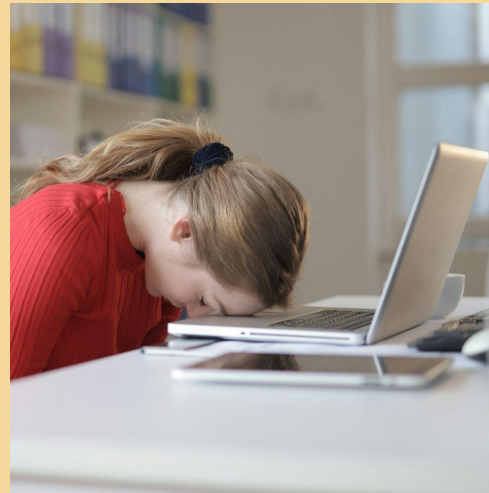
By Bhekisisa.org

Is there a difference in average test scores among students from different teaching methods (traditional, online, hybrid)?



By [Jimmy Chin](http://JimmyChin.com)

How does the amount of daily physical activity (measured in hours) predict the likelihood of experiencing anxiety symptoms (yes, no)?



What is the relationship between hours of sleep per night and cognitive function (measured by memory recall)?

How do we go from here?

Independent	
Dependent ¹	
	
	Chi-squared test
	 Student's <i>t</i> test One-way ANOVA Factorial ANOVA
	Simple regression Multiple regression

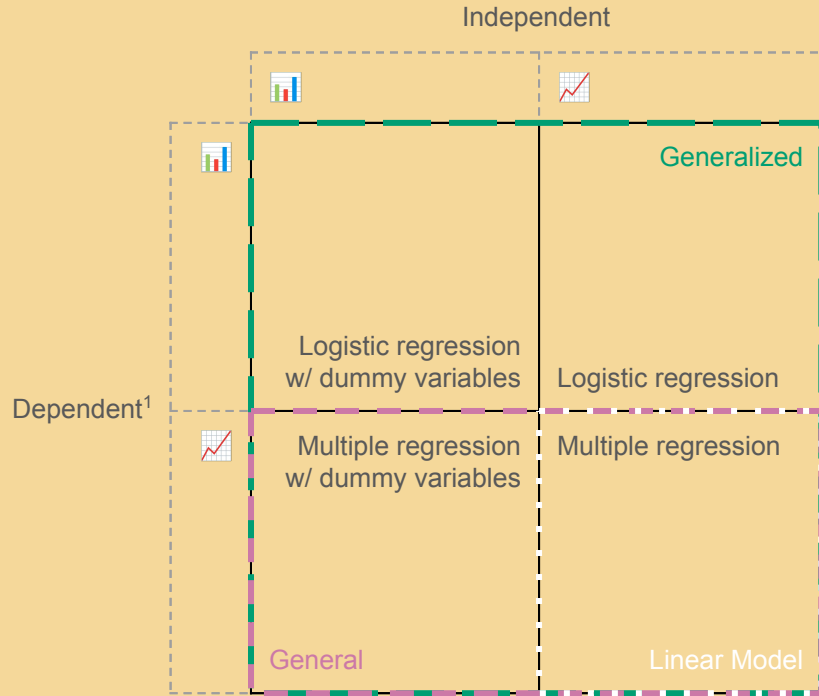
From one predictor to multiple predictors



Categorical: dichotomous¹, polytomous

Continuous: real¹, integer, bounded (e.g., 0 to 1)

How do we go from here?

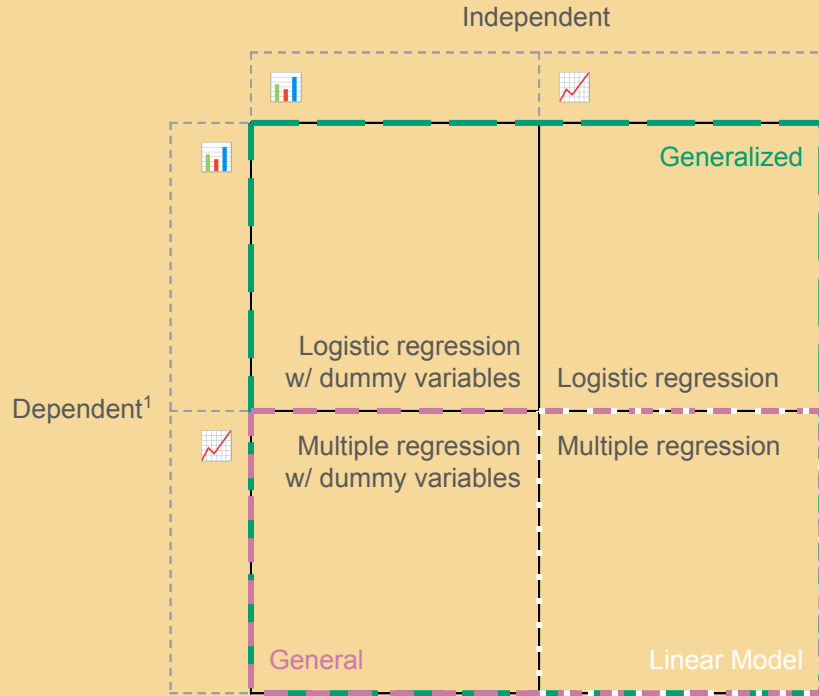


- ❑ From one predictor to multiple predictors
- ❑ From a toolbox to a multitool (“Swiss Army knife”)

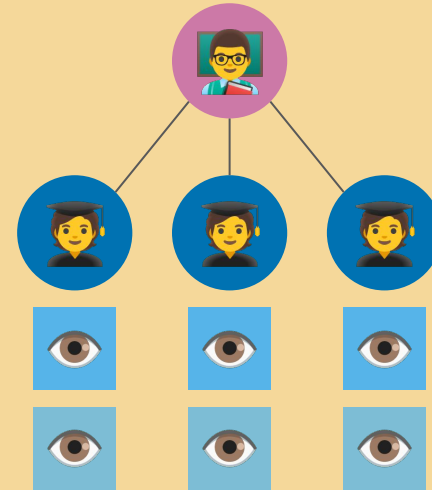


 Categorical: dichotomous¹, polytomous
 Continuous: real¹, integer, bounded (e.g., 0 to 1)

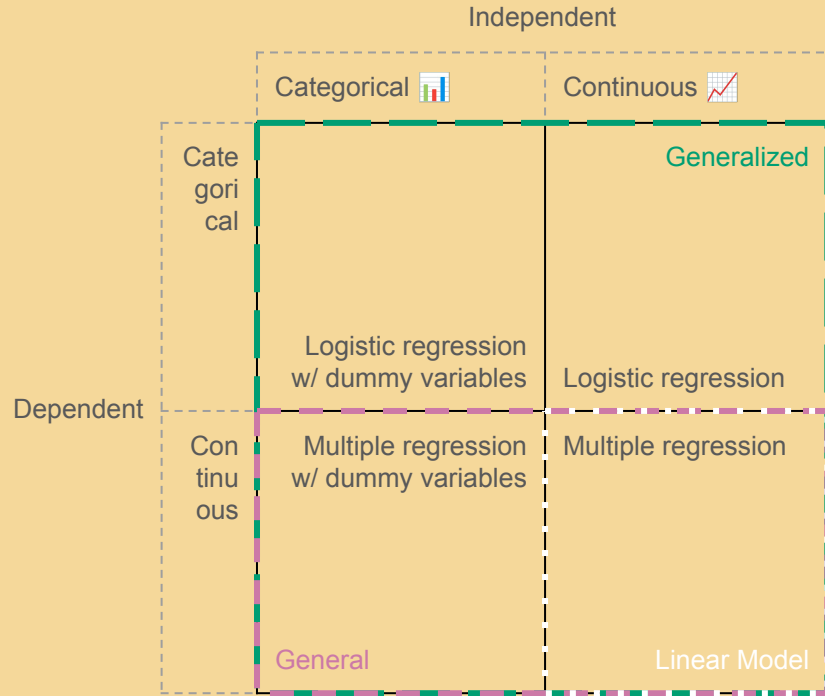
How do we go from here?



- From one predictor to multiple predictors
- From a toolbox to a multitool
- Hierarchical data analysis



How do we go from here?



- From one predictor to multiple predictors
- From a toolbox to a multitool
- Hierarchical data analysis
-  Bayesian inference (vs. frequentist inference)

Course overview



Lectures

[Alexander Savi](#) (he/him)

- ❑ Honours BSc Psychology
- ❑ Research MSc Psychology
- ❑ PhD Psychological Methods
- ❑ Postdoc Learning Analytics
- ❑ Assistant Professor of Psychological Methods

Topics

1. Statistical reasoning with GLM
2. Multiple linear regression (GLM)
3. Dummy-variable regression (GLM)
4. Intermezzo: Bayesian statistics
5. Logistic regression (GLM)
6. Multilevel and longitudinal analysis (GLM)
7. Statistics superpowers



Questions during & after lecture

Why attend

- ❑ Prioritized concepts from the book
- ❑ Fresh examples
- ❑ Bare necessary R code
- ❑ Book links for quick reference
- ❑ Topics not covered by the book
- ❑ Some extras for those who want more
- ❑ Ask questions

Assistants



Miró the Superhero



Willem the Supervillain



Web lectures can fail, use for re-viewing



Weekly assignments*

Jonas van Nijnatten (he/him) + TAs

- ☐ Four sections
 - ☐ R & Tidyverse
 - ☐ Begrip & Inzicht
 - ☐ Analyse & Interpretatie
 - ☐ Casus
- ☐ Formative
- ☐ Q&A session
 - ☐ prepare questions or use it to work on it
- ☐ Use decimal points (.)
- ☐ Great exam preparation
- ☐ Help improve assignments

 Online Q&A session (drop-in):
(almost) every Friday 10:00–10:45

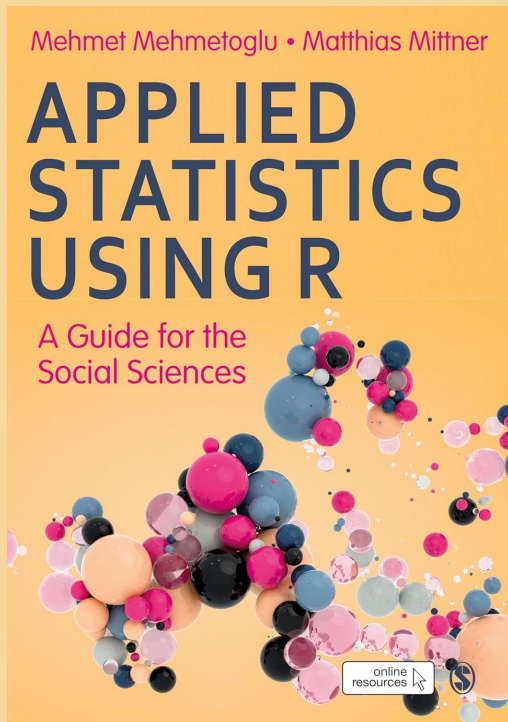
Grading

- ☐ Every Saturday 09:00 to Friday 23:55
- ☐ 80% correct is sufficient, 70% correct depends on shown effort
- ☐ 7/7 sufficient
- ☐ Two attempts per assignment
- ☐ Two catch-up weeks
 - ☐ If sufficient after first: correction to grade
 - ☐ If sufficient after second: correction to retake
- ☐ No feedback; correct answers will be available for a week





Book



Pros

- ❑ Builds on Chapter 18 of Whitlock & Schluter (where you left of last year)
- ❑ Not only for social sciences
- ❑ Modern approach (GLM and Tidyverse)
- ❑ Frequentist and Bayesian inference, with examples and code
- ❑ Good reference during the remainder of your study
- ❑ It's a book!

Cons

- ❑ It's a book!
- ❑ Some students find it not easy

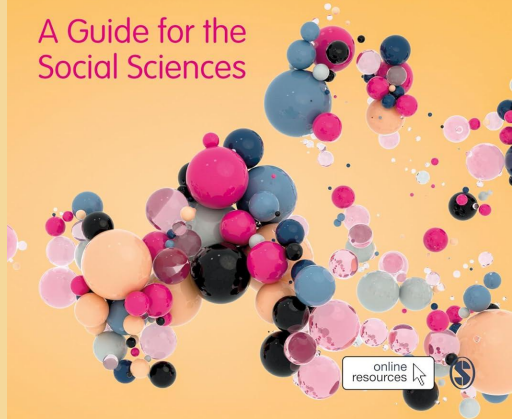


Book

Mehmet Mehmetoglu • Matthias Mittner

APPLIED STATISTICS USING R

A Guide for the
Social Sciences



Active reading

- ❑ Use [skimming](#) for some chapters, so you can scan them when there's something you really need to know.
- ❑ Use [intensive reading](#) (or 'study') for others (check [online resources](#) for executing R code).

Prepare the literature before the start of the week.

100 Exam*

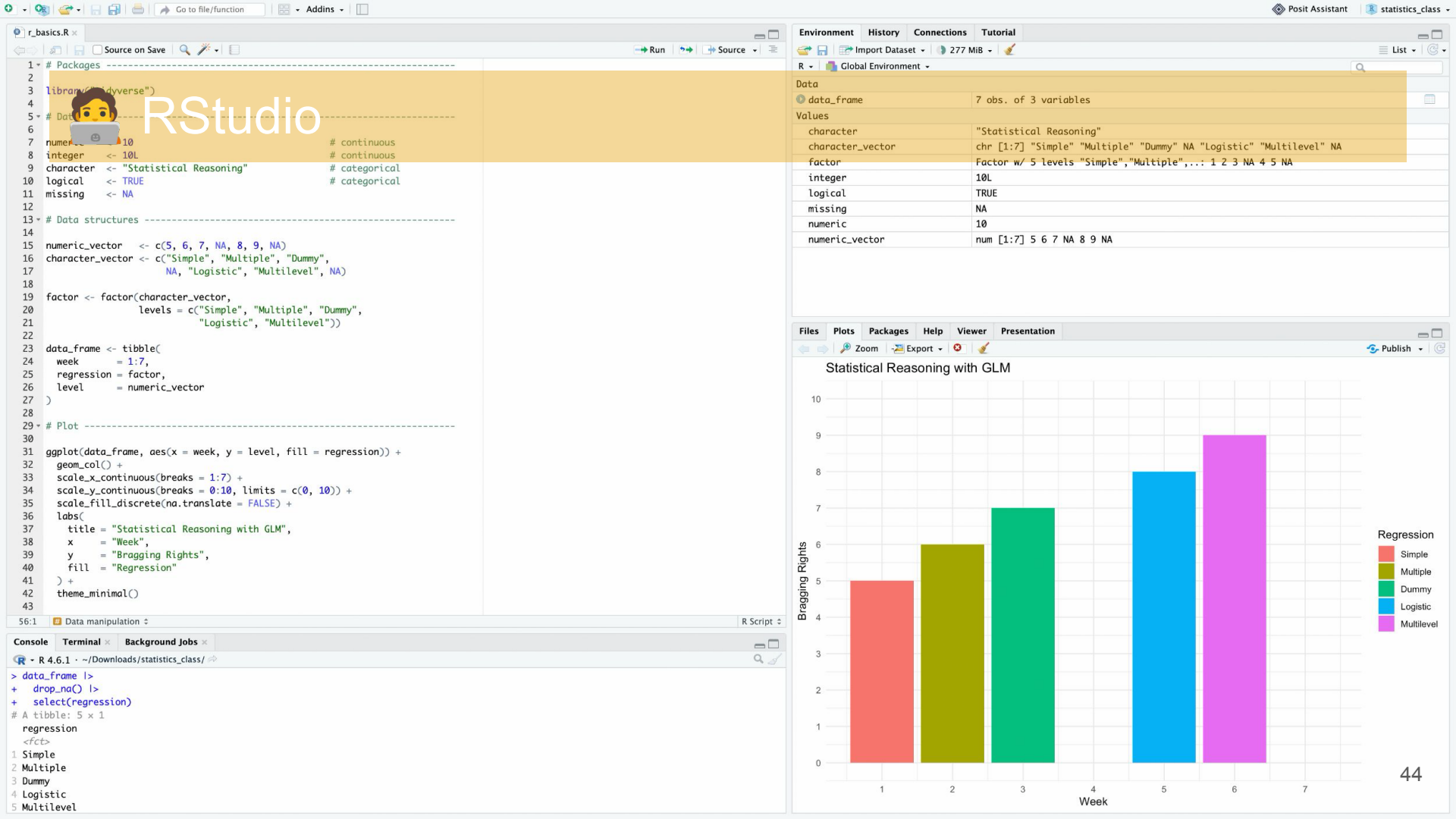
- ❑ 5/8 SR
- ❑ Digital open book (⚠)
- ❑ Practice makes perfect
 - ❑ weekly assignments *Begrip & Inzicht* and *Analyse & Interpretatie* sections
 - ❑ exam(ple) questions from previous years
- ❑ Learning goals (studiewijzer)
- ❑ Always do the exam
- ❑ Exam review: Friday, November 13, 9–12



Statistics for all: open-source pioneers win million-dollar prize

The free, open-source software R has fundamentally transformed research in statistics and data science — and continues to shape the field to this day. Now the pioneers who have driven its development for some 30 years are receiving the Rousseeuw Prize for Statistics,

— [ETH](#) (Jun. 17, 2026)



“Keep the robots out of the gym”



[Daniel Miessler](#); [Luke Shepard](#); [Bruce Schneier](#)



[UvA AI guide](#)



Today

Topics

- 1 | Statistical reasoning with GLM
 - 1.1 |  Simple linear regression
 - 1.2 | Simulation superpower
- 2 | Multiple linear regression
- 3 | Dummy-variable regression
- 4 | Bayesian statistics (a brief introduction)
- 5 | Logistic regression
- 6 | Multilevel and longitudinal analysis
- 7 | Statistics superpowers

Learning goals

Revisit simple linear regression, which is at the core of this course.

Get acquainted with statistical simulations and see how they can help elucidate statistical concepts.

Revisit some of the concepts of MvO1.

Simple linear regression

IJskoud de Beste

Joop

Can I use the weather forecast to predict the amount of ice cream to make?

What is the relationship between temperature and ice cream sales?

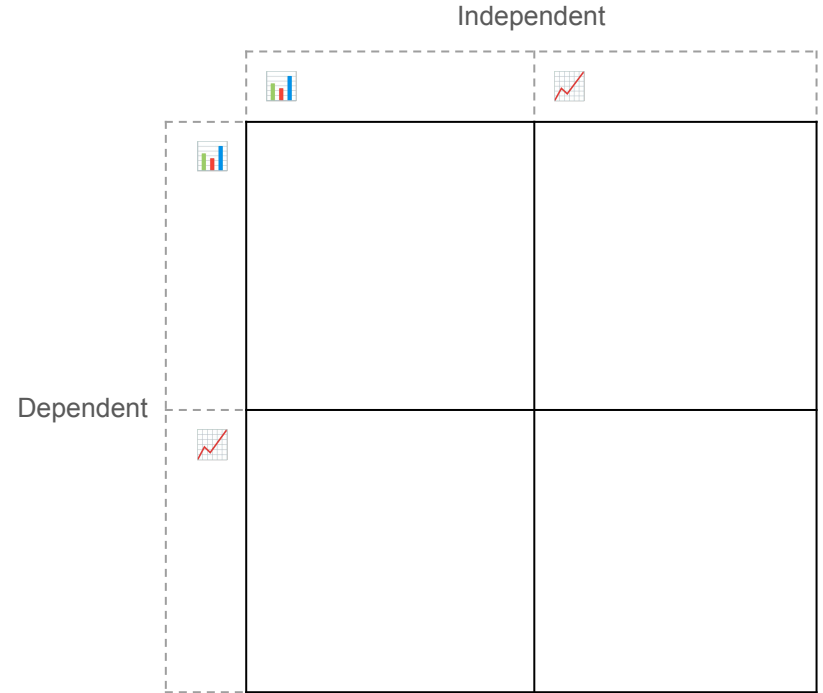
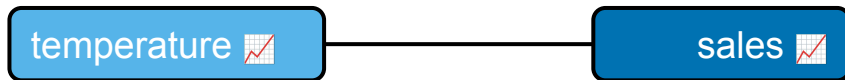
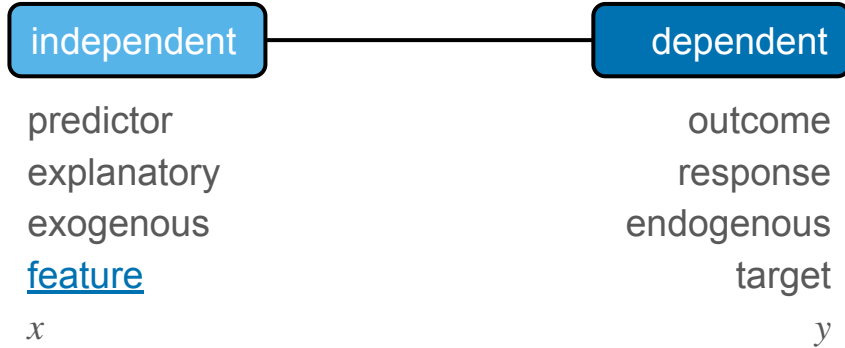
Me

temperature 

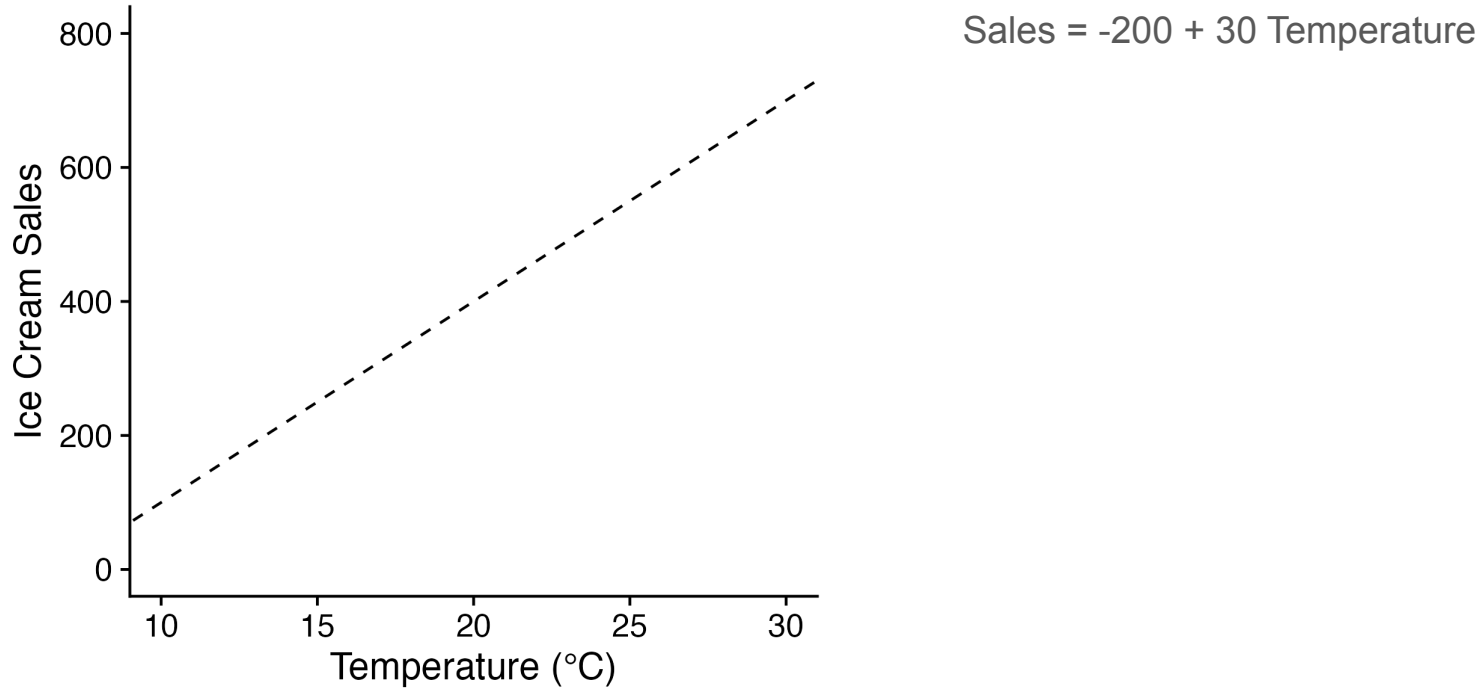
sales 



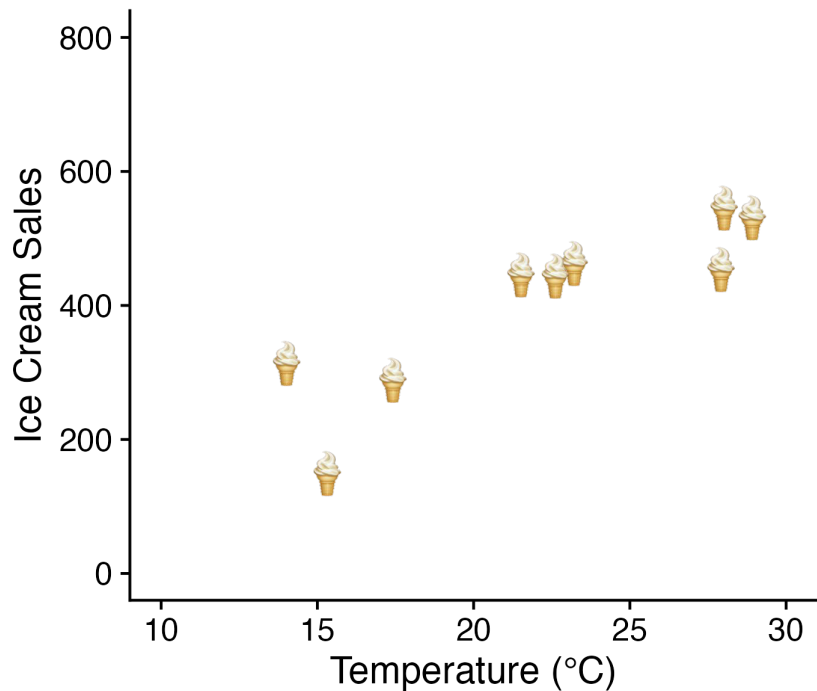
Conceptual model | Relationships between variables



Functional form | Linear relationships between variables



Data collection

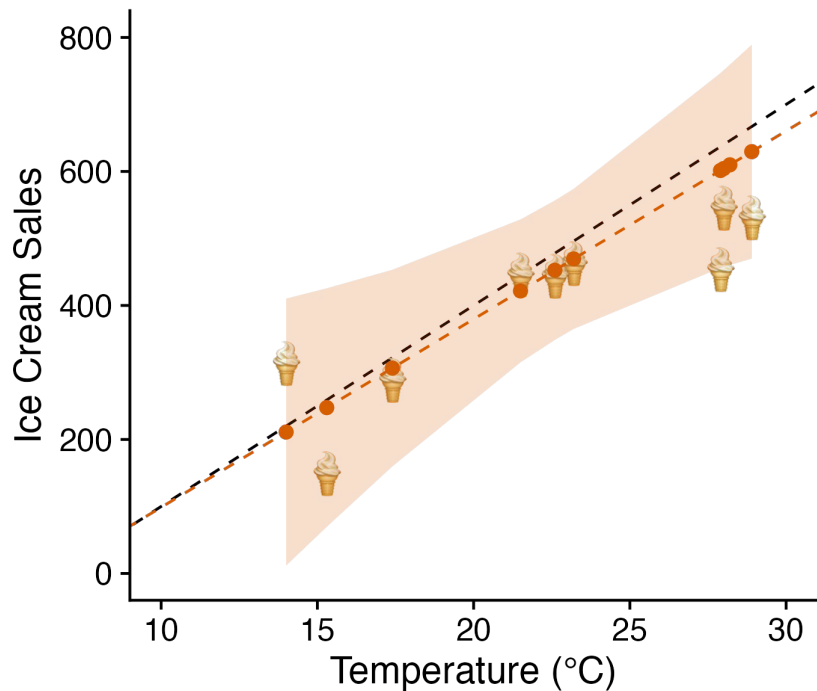


```
ice_cream_data <- tibble::tribble(  
  ~temperature, ~ice_cream_sales,  
  27.9, 452,  
  15.3, 148,  
  17.4, 287,  
  21.5, 444,  
  28.2, 935,  
  14, 312,  
  28, 544,  
  28.9, 529,  
  23.2, 461,  
  22.6, 442)
```

$$\text{ice_cream_sales} = \beta_0 + \beta_1(\text{temperature}) + \epsilon$$

```
mod <- ice_cream_sales ~ temperature# y ~ x  
mod <- ice_cream_sales ~ 1 + temperature# explicit  
intercept
```

Modeling | Linear regression



```
fit <- lm(mod, data = ice_cream_data)
summary(fit); confint(fit, level = .95)
```

Call:

```
lm(formula = model, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-149.46	-89.73	-15.06	14.62	325.12

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-182.199	197.447	-0.923	0.3831
temperature	28.088	8.468	3.317	0.0106 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 142.7 on 8 degrees of freedom

Multiple R-squared: 0.579, Adjusted R-squared: 0.5264

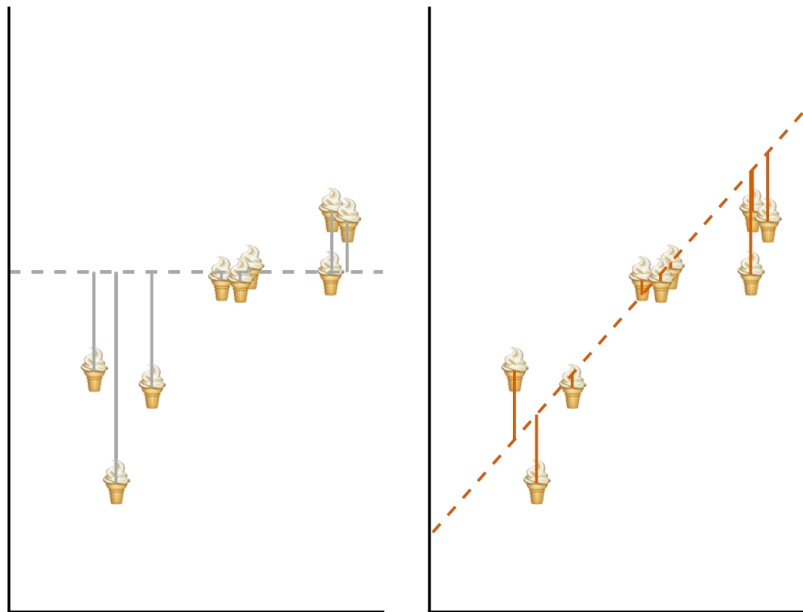
F-statistic: 11 on 1 and 8 DF, p-value: 0.01059

For every °C increase, sales increase on average with 9 to 48 ice creams.

$$\widehat{\text{ice_cream_sales}} = -182.2 + 28.09(\text{temperature})$$

Modeling | Ordinary least squares

Residual SS ($\beta_1 = 0$) Residual SS ($\beta_1 = 28$)



Objective: minimize residual sums of squares (RSS)

Baseline RSS $Sales_i = \beta_0 + e_i$: 387052.4

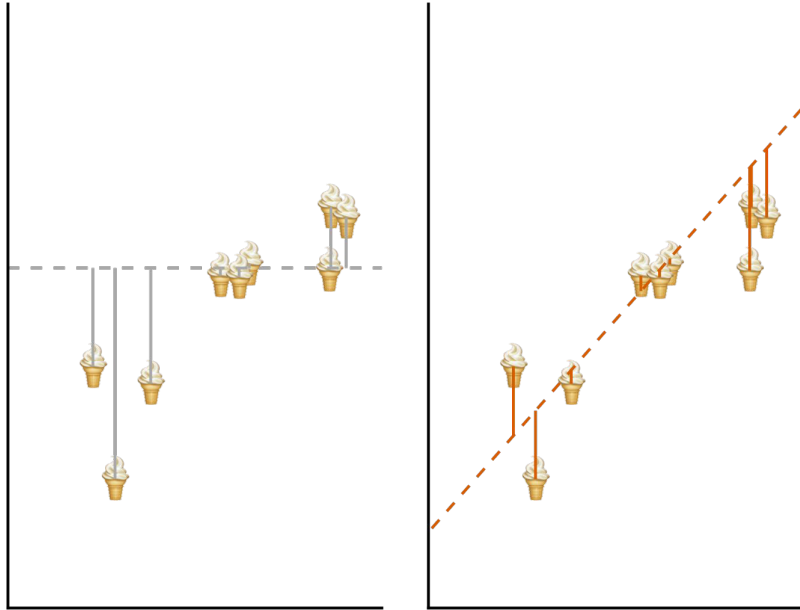
Model RSS $Sales_i = \beta_0 + \beta_1 Temperature_i + e_i$: 162946

```
mod_baseline <- ice_cream_sales ~ 1
fit_baseline <- lm(mod_baseline, data = data)
summary(fit_baseline)

deviance(fit)
deviance(fit_baseline)
```

Evaluation | How well does the model describe the data?

Residual SS ($\beta_1 = 0$) Residual SS ($\beta_1 = 28$)



On average, how far off is my regression estimate from the observed values?

Standard deviation of the residuals, or root mean squared error (RMSE): $\sqrt{(RSS / (n - K))}$

Model RMSE: 142.717 ice creams off

Is this good or bad?

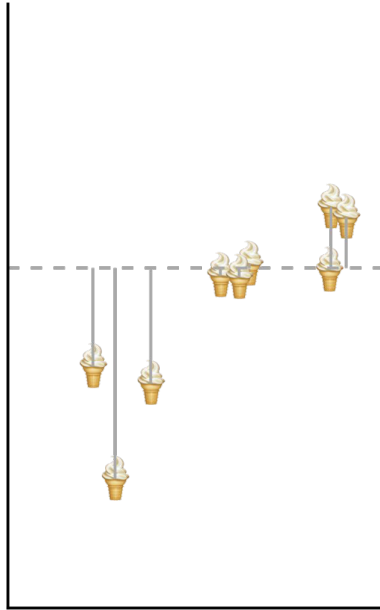
Baseline RMSE: 207.379

Still, is it good or bad? Prediction!

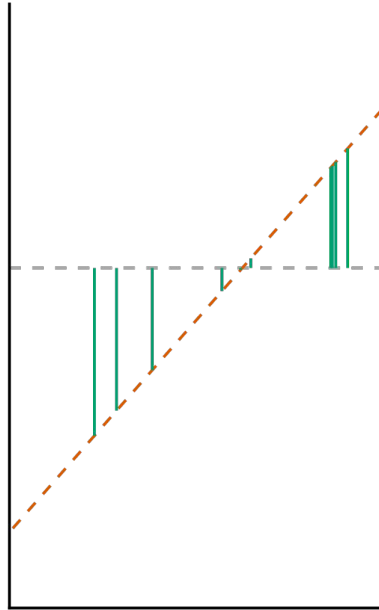
```
summary(fit)$sigma  
summary(fit_baseline)$sigma
```

Evaluation | How well does the model describe the data?

Total SS



Explained SS



What is the proportion of variance shared by the data and the model? ('verklaard' only if experimental!)

Coefficient of determination (R^2), or 'explained' variance: ESS / TSS

Model R^2 : 0.579

58% of variation in sales is shared with the model (i.e., temperature)

$R^2 = 1$ (model fits data perfectly)

$R^2 = 0$ (model doesn't do better than mean)

```
summary_fit$r.squared
```

Evaluation | Is the relationship statistically significant?

Global significance (full model): F test

$$R^2 = 0.579 \text{ (fitted)}$$

$$H_0: R^2 = 0$$

$$H_A: R^2 > 0$$

See next lectures.

Local significance (one coefficient): t value

$$\beta_1 = 28.088 \text{ (fitted)}$$

$$H_0: \beta_1 = 0$$

$$H_A: \beta_1 > 0$$

See simulation superpower.

💡 Compute t -statistic for β_1 (same procedure as for the mean): $t = (28.088 - 0) / 8.468 = 3.317$

Evaluation | What is the size of the effect?

Global effect size (full model)

R^2 (correlation family, variance explained)

- ❑ $R^2 \geq .02$ weak; .13 moderate; .26 substantial; Cohen (1988)

Cohen's f^2 (correlation family, variance explained)

- ❑ $f^2 = R^2 / (1 - R^2) = \text{explained} / \text{unexplained}$
- ❑ $f^2 \geq .02$ small; .15 medium; .35 large; Cohen (1988)

Local effect size (one coefficient)

Cohen's f^2 (correlation family, variance explained)

- ❑ $f^2 = (R^2_{AB} - R^2_A) / (1 - R^2_{AB})$
- ❑ No known rules of thumb

Rules of 👍: arbitrary conventions when no substantiated rules exist.

Evaluation | Sign, size, significance, smelt

💡 For every °C increase, sales increase on average with 9 to 48 ice creams.

💡 On average, the model estimate is 142.717 ice creams off from the observed values.

💡 58% of variation in sales can be attributed to the model; $F(1, 8) = 11.003$, $p = .011$ (alpha = .05).

💡 The slope of temperature differed significantly from zero; $t(8) = 3.50$, $p = .011$ (two-tailed, alpha = .05).

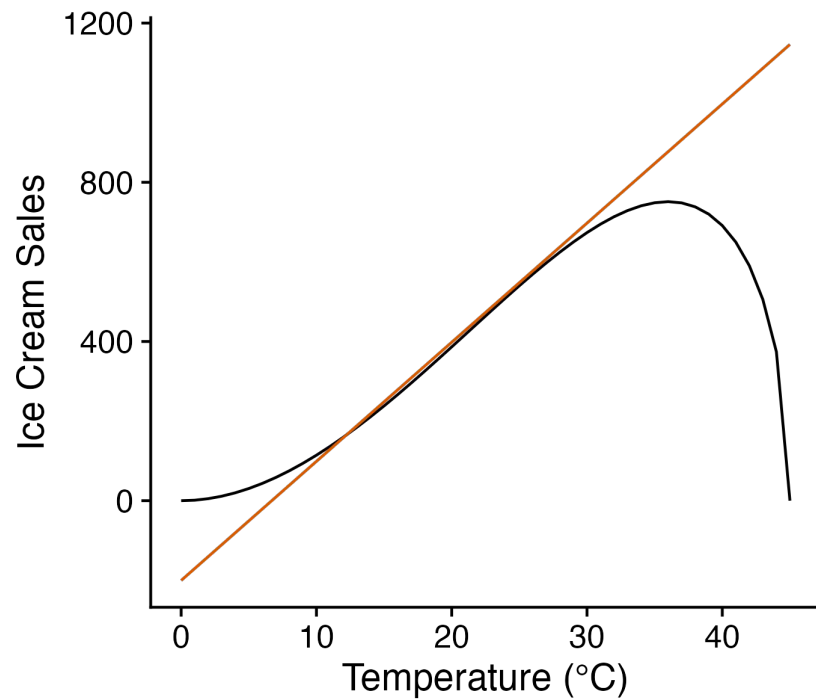
💡 Effect size f^2 is 1.375, which is considered large (Cohen, 1988).



Prediction

- ❑ Linearity assumption
- ❑ Evaluate model on new data

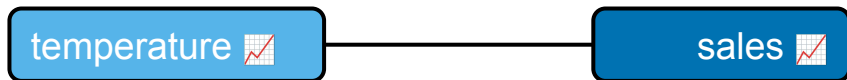
```
new_data <- tibble(temperature = c(-10, 0, 10, 20,  
30, 100))  
predict(fit, newdata = new_data)
```



Model thinking

What is the relationship between temperature and ice cream sales?

Conceptual model



Linear regression equation

$$\text{ice_cream_sales} = \beta_0 + \beta_1(\text{temperature}) + \epsilon$$

R formula

```
mod <- ice_cream_sales ~ temperature
```

Introduction to R

Curled by Danielle Navarro ([jasmines package](#), [diy](#)) 🎨

Environment History Connections Tutorial

Import Dataset 277 MiB

R - Global Environment

Data

data_frame 7 obs. of 3 variables

Values

character	"Statistical Reasoning"
character_vector	chr [1:7] "Simple" "Multiple" "Dummy" NA "Logistic" "Multilevel" NA
factor	Factor w/ 5 levels "Simple","Multiple",...: 1 2 3 NA 4 5 NA
integer	10L
logical	TRUE
missing	NA
numeric	10
numeric_vector	num [1:7] 5 6 7 NA 8 9 NA

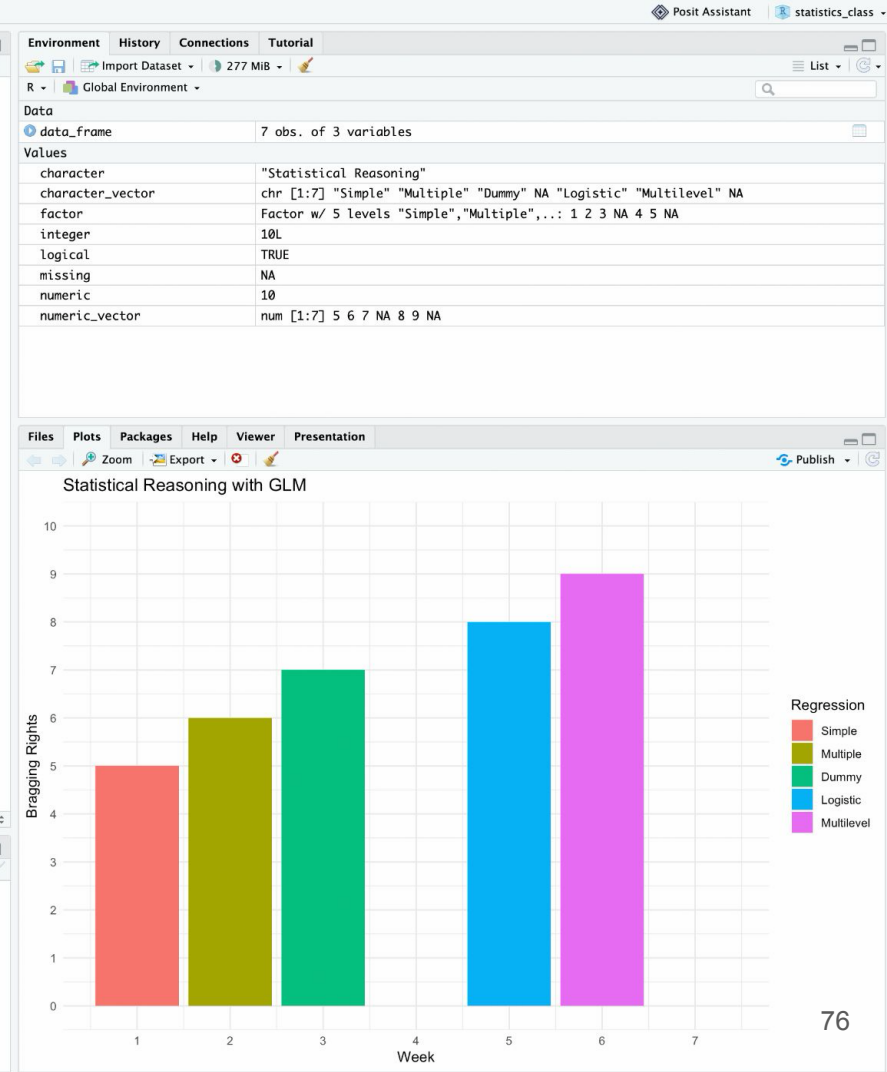
```
1 # Packages
2
3 library("tidyverse")
4
5 # Data types -----
6
7 numeric <- 10 # continuous
8 integer <- 10L # continuous
9 character <- "Statistical Reasoning" # categorical
10 logical <- TRUE # categorical
11 missing <- NA
12
13 # Data structures -----
14
15 numeric_vector <- c(5, 6, 7, NA, 8, 9, NA)
16 character_vector <- c("Simple", "Multiple", "Dummy",
17                       NA, "Logistic", "Multilevel", NA)
18
19 factor <- factor(character_vector,
20                 levels = c("Simple", "Multiple", "Dummy",
21                           "Logistic", "Multilevel"))
22
23 data_frame <- tibble(
24   week = 1:7,
25   regression = factor,
26   level = numeric_vector
27 )
28
29 # Plot -----
30
31 ggplot(data_frame, aes(x = week, y = level, fill = regression)) +
32   geom_col() +
33   scale_x_continuous(breaks = 1:7) +
34   scale_y_continuous(breaks = 0:10, limits = c(0, 10)) +
35   scale_fill_discrete(na.translate = FALSE) +
36   labs(
37     title = "Statistical Reasoning with GLM",
38     x = "Week",
39     y = "Bragging Rights",
40     fill = "Regression"
41   ) +
42   theme_minimal()
43
```

56:1 Data manipulation R Script

Console Terminal Background Jobs

R - R 4.6.1 - ~/Downloads/statistics_class

```
> data_frame |>
+ drop_na() |>
+ select(regression)
# A tibble: 5 x 1
  regression
  <fct>
1 Simple
2 Multiple
3 Dummy
4 Logistic
5 Multilevel
```



Best practices

- ❑ Create a [new project](#) for this course
- ❑ Use scripts and save regularly
 - ❑ .R
 - ❑ .Rmd ([R Markdown](#))
 - ❑ .qmd ([Quarto](#))
- ❑ Use [comments](#) excessively
- ❑ Clear history
- ❑ Understand pipes
 - ❑ `function(function(x))`
 - ❑ `x |> function() |> function()`
 - ❑ `x %>% function() %>% function()`

Help with installing packages? Setting up environment, also see book

Remind of script that comes with the book.



Cooling Down

Curled by Danielle Navarro ([jasmynes package](#), [diy](#)) 🎨



What's next



Weekly assignment

- I. Attempt problem independently
 - Consult official resources (literature, lectures)
 - This is what will be examined
- II. Consult fellow students or teachers
 - Collaborate and discuss with peers
 - Attend lecture & ask questions
 - Join the weekly drop-in consultation
 - Active engagement will increase your success
- III. Use [UvA AI chat](#), responsibly
 - Previous students found answers confusing
 - Only if I & II are satisfied
 - Treat as a coach, not as a solver
 - Verify and reflect, beware of plagiarism
 - Prediction $\neq$ intelligence

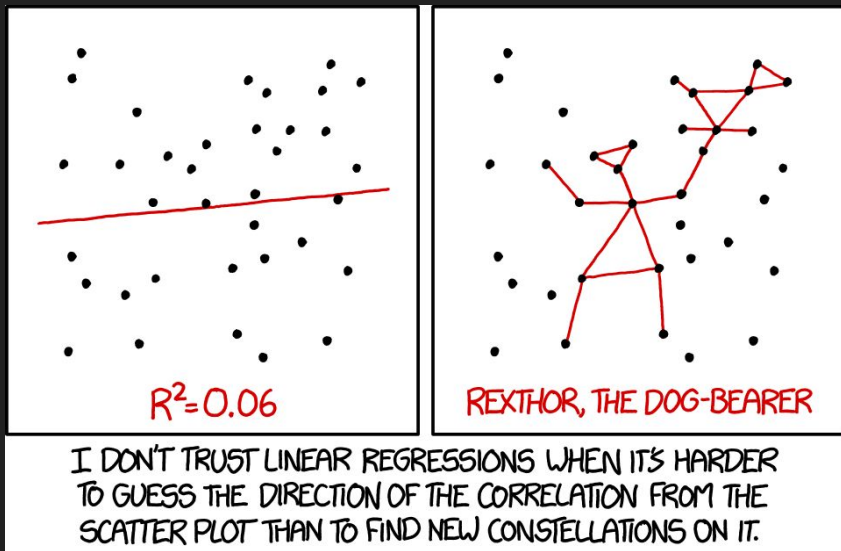


Next week



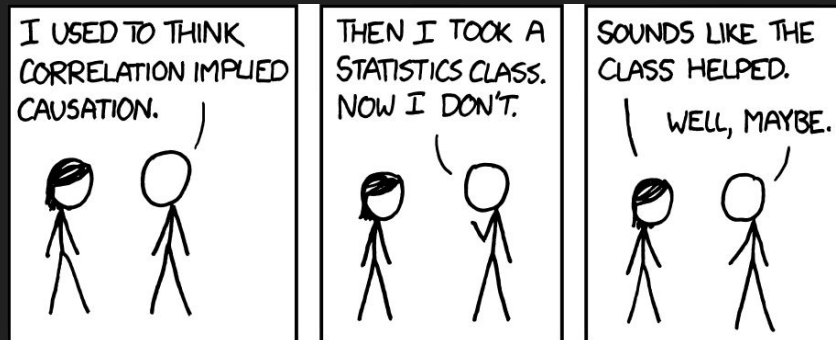
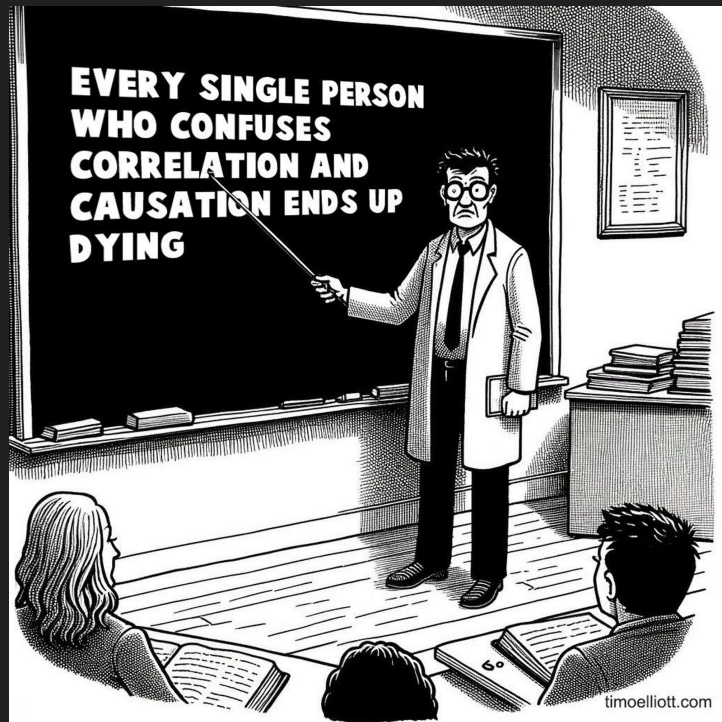


🤔 Get it? | Correlation

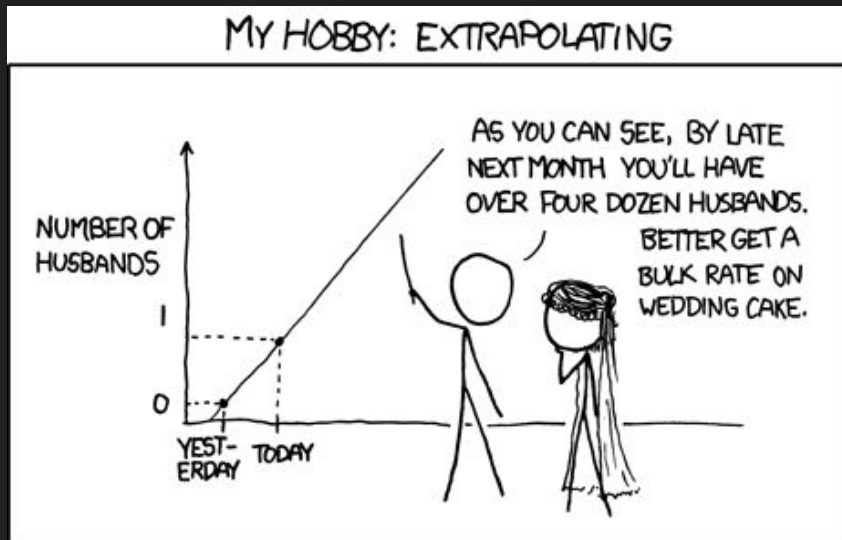




Get it? | Correlation does not imply causation



🤔 Get it? | Extrapolation





Colophon

Slides

alexandersavi.nl/teaching/

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Approved for [Free Cultural Works](https://freeculturalworks.org/).